

DATA 266 - Homework 3 Findings

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This report summarizes twelve code-driven prompt-engineering experiments and two separately trained single-head attention models implemented from scratch.

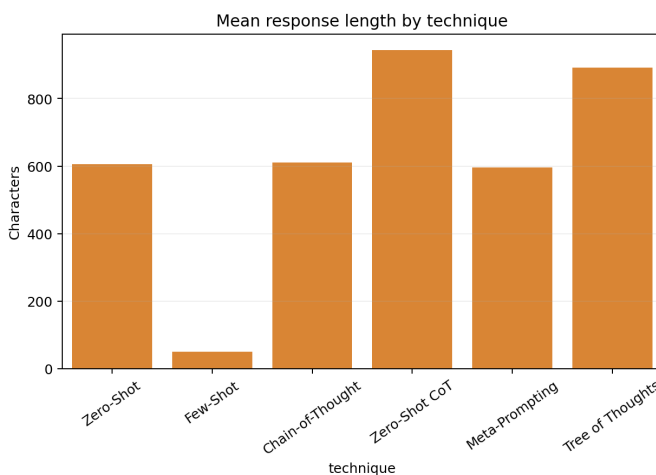
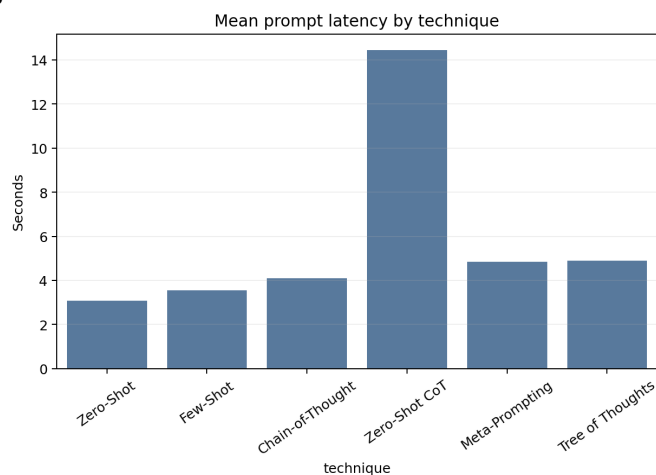
1. Prompt Engineering

All prompts were called through LangChain using gemini-3.6-flash with temperature zero. Each technique was tested on the same math and scheduling-logic tasks. The automated term check verifies whether each response contains the expected answer elements; the full responses remain in the notebook and JSON output for qualitative inspection.

Technique	Examples	Mean latency (s)	Mean chars	Term-check rat
Zero-Shot	2	3.069	605.5	100%
Few-Shot	2	3.561	50.0	100%
Chain-of-Thought	2	4.091	610.0	100%
Zero-Shot CoT	2	14.446	943.5	50%
Meta-Prompting	2	4.836	595.5	50%
Tree of Thoughts	2	4.890	892.5	50%

Comparison findings

In this run, Few-Shot produced the shortest responses on average, Zero-Shot CoT produced the longest, and Zero-Shot had the lowest mean latency. Zero-shot prompting was the least constrained. Few-shot prompting demonstrated the output format. The CoT variants made equations and checks visible. Meta-prompting added an extra prompt-design stage, while Tree of Thoughts explicitly generated and compared alternatives. The results show that additional structure changes response detail and verification behavior, but greater length alone does not guarantee correctness.



Representative generated outputs

Zero-Shot - Example 1 (Math)

To find the number of adult and child tickets sold, we can set up a system of linear equations.

Let a be the number of adult tickets and c be the number of child tickets.

- 1) $a + c = 120$
- 2) $12a + 7c = 1165$

From equation (1), we can express c as $c = 120 - a$. Substitute this into equation (2):

$$\begin{aligned} 12a + 7(120 - a) &= 1165 \\ 12a + 840 - 7a &= 1165 \\ 5a + 840 &= 1165 \\ 5a &= 325 \\ a &= 65 \end{aligned}$$

Now, find c :

$$c = 120 - 65 = 55$$

Final Answer:

The theater sold **65 adult tickets** and **55 child tickets**.

Zero-Shot - Example 2 (Logic)

Based on the given constraints, we can determine the schedule step-by-step:

1. **Thursday:** **Cara** presents on Thursday (given).
2. **Tuesday & Wednesday:** Dev presents immediately after Ben, forming a two-day block ([Ben, Dev]). Since Ben cannot present on Monday and Thursday is taken by Cara, the only available two-day block for Ben and Dev is Tuesday and Wednesday. Therefore, **Ben** presents on Tuesday and **Dev** presents on Wednesday.
3. **Monday:** The only remaining person and day is **Ava** on Monday. (This also satisfies the rule that Ava presents before Cara).

Complete Schedule:

* **Monday:** Ava
* **Tuesday:** Ben
* **We...**

Few-Shot - Example 1 (Math)

65 adult tickets and 55 child tickets.

Few-Shot - Example 2 (Logic)

Monday - Ava; Tuesday - Ben; Wednesday - Dev; Thursday - Cara.

Chain-of-Thought - Example 1 (Math)

****Reasoning summary:****

Let $a + c = 120$ and $12a + 7c = 1,165$, where a represents the number of adult tickets and c represents the number of child tickets.

Substitute $c = 120 - a$ into the revenue equation:

$$12a + 7(120 - a) = 1,165$$

$$12a + 840 - 7a = 1,165$$

$$5a + 840 = 1,165$$

$$5a = 325$$

$$a = 65$$

Now, solve for c :

$$c = 120 - 65 = 55$$

****Verification:****

$$\text{* Total tickets: } 65 + 55 = 120$$

$$\text{* Total revenue: } 12(65) + 7(55) = 780 + 385 = 1,165$$

****Final answer:**** 65 adult tickets and 55 child tickets.

Chain-of-Thought - Example 2 (Logic)

****Fixed constraint:****

Cara presents Thursday.

****Eliminate impossible placements:****

With Cara assigned to Thursday, the remaining days to fill are Monday, Tuesday, and Wednesday for Ava, Ben, and Dev. Since Dev presents immediately after Ben, Ben and Dev must occupy consecutive days—either (Monday, Tuesday) or (Tuesday, Wednesday). Because Ben cannot present on Monday, the (Monday, Tuesday) placement is eliminated. Thus, Ben must present on Tuesday and Dev on Wednesday. This leaves Monday as the only remaining slot for Ava.

****Verify every rule:****

1. *Ava presents before Cara: * Ava is on Monday and Cara is on Thursday (Rule verified).

2. *B...

Zero-Shot CoT - Example 1 (Math)

Here is the step-by-step solution to the problem:

Step 1: Define Variables

* Let a = the number of adult tickets sold.

* Let c = the number of child tickets sold.

Step 2: Form Equations

Based on the problem statement, we can set up two equations:

1. **Ticket count equation:**

$$a + c = 120$$

2. **Revenue equation:**

$$12a + 7c = 1,165$$

Step 3: Solve the Equations

We can solve this system using substitution.

1. Rearrange the first equation to express c in terms of

Zero-Shot CoT - Example 2 (Logic)

To determine the complete schedule, we apply the constraints step by step:

1. **Thursday:** Constraint 4 states that **Cara** presents on Thursday.

* **Remaining days:** Monday, Tuesday, Wednesday.

* **Remaining people:** Ava, Ben, Dev.

2. **Tuesday & Wednesday:** Constraint 3 states that **Dev** presents immediately after **Ben**, which means Ben and Dev must occupy two consecutive days ([Ben, Dev]).

* The available consecutive days in the remaining slot (Mon, Tue, Wed) are either (Mon, Tue) or (Tue, Wed).

* Constraint 2 states that **Ben** cannot present on Monday.

* Therefore, the pair [Ben, Dev] must be placed on **Tues...**

Meta-Prompting - Example 1 (Math)

Here is the step-by-step solution to the problem:

Step 1: Define the Variables

* Let A = the number of adult tickets sold

* Let C = the number of child tickets sold

Step 2: Set Up the System of Equations

Based on the information given:

1. **Total tickets sold:**

$$A + C = 120$$

2. **Total revenue:**

$$12A + 7C = 1,165$$

Step 3: Solve the System of Equations

We can use the substitution method:

1. Express C in terms of A using the first equation:

$$C = 120 - A$$

2. Substitute this expression for C into the revenue equation:

$$12A + 7(120 - A) = 1,165$$

3. Distribute and simpl...

Meta-Prompting - Example 2 (Logic)

STEP 1: CONSTRAINTS

* **Entities:** Ava (A), Ben (B), Cara (C), Dev (D)

* **Days:** Monday,

Tree of Thoughts - Example 1 (Math)

Here is the solution structured using the compact tree-of-thought procedure.

Step 1: Generate Candidate Approaches

* **Branch 1 (Substitution):** Express C in terms of A using total tickets ($C = 120 - A$) and substitute into the revenue equation.

* **Branch 2 (Elimination):** Scale the ticket count equation by 7 to eliminate C when subtracted from the revenue equation.

* **Branch 3 (Bounded Integer Checking):** Start at the baseline bound where all tickets are child tickets and adjust for the price differential.

Tree of Thoughts - Example 2 (Logic)

Here is the step-by-step solution using the Tree-of-Thought procedure:

Step 1: Create Candidate Branches for Ben–Dev Pairs

Since Dev presents immediately after Ben, the possible consecutive (Ben, Dev) day pairs are:

* **Branch A:** Ben = Monday, Dev = Tuesday

* **Branch B:** Ben = Tuesday, Dev = Wednesday

* **Branch C:** Ben = Wednesday, Dev = Thursday

Step 2: Prune Branches Based on Constraints

* **Branch A:** **Pruned** — Violates the constraint "Ben does not present Monday."

* **Branch C:** **Pruned** — Violates the constraint "Cara presents Thursday" (Dev would conflict with Cara on Thursday).

* **Branch B:** **...

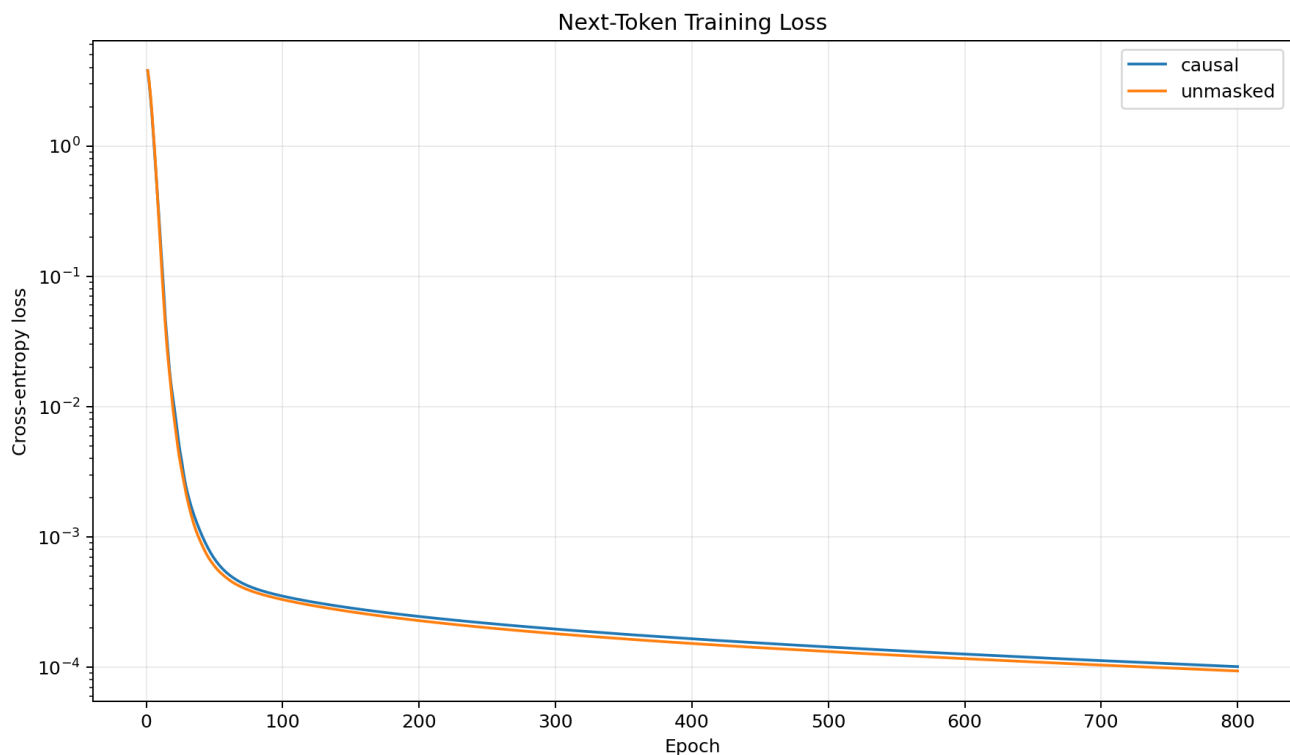
2. Learned Self-Attention

The exact assignment text was tokenized at the word level. Each model learned token and position embeddings plus W_q , W_k , W_v , W_o , and a vocabulary output head. The attention scores were computed explicitly as QK^T divided by $\sqrt{d_k}$, followed by softmax and multiplication by V . Both models used the same seeded initialization and next-token objective, but they were trained as separate model instances and saved as separate checkpoints.

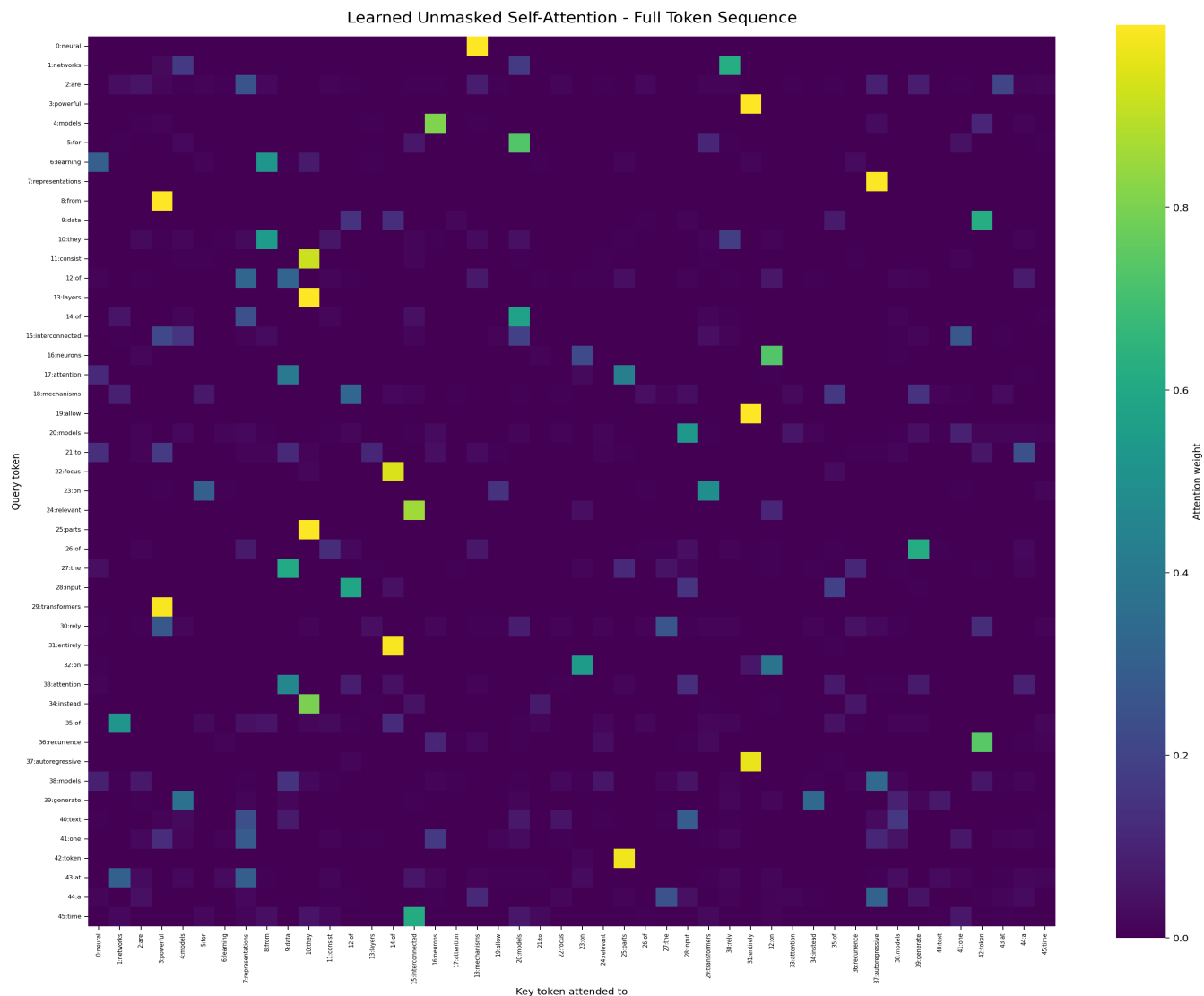
Model	Mask	Final loss	Accuracy	Train s	Future mass	Max future we
unmasked	False	0.000094	100.0%	4.480	17.971975	0.99999344
causal	True	0.000101	100.0%	3.048	0.000000	0.00000000

Attention findings

The unmasked model assigned 17.971975 total attention mass above the diagonal. The causal model assigned 0.0000000000, with a maximum individual future-token weight of 0.0000000000. The zero upper triangle verifies that the mask was applied before softmax and prevents every token from attending to future positions. The loss curves confirm that both heatmaps were produced after learning rather than from random initialization.

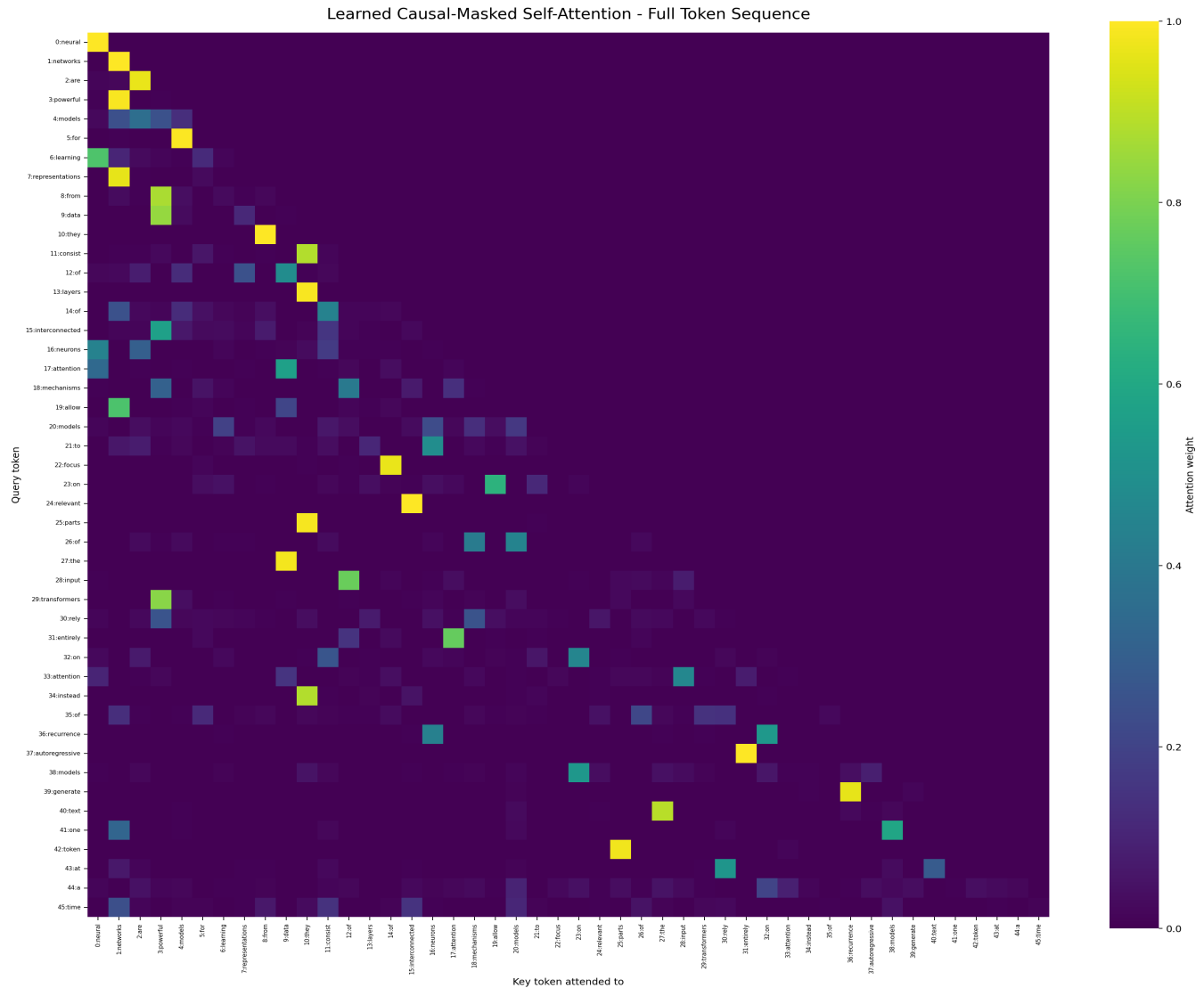


Unmasked learned attention



Nonzero values are permitted on both sides of the diagonal because no causal restriction is applied to the unmasked model.

Causal-masked learned attention



The visibly empty upper triangle shows that query positions cannot attend to future key positions. The numerical assertion independently verifies this property.

3. Conclusion

Prompt design changed the amount of structure, verification, and computational overhead in the generated responses. Few-shot examples improved format control, CoT prompts emphasized auditable calculations, meta-prompting separated instruction design from execution, and Tree of Thoughts compared candidate paths. In the attention experiment, both models learned from the same exact text and objective, but only the causal model enforced autoregressive information flow. Its zero future attention mass is the central evidence that causal masking worked.

References

Vaswani et al. (2017), Attention Is All You Need, Section 3.2. Google AI for Developers, Gemini model documentation. LangChain Python reference for ChatGoogleGenerativeAI.